

✓ Multimedia Homework 6-I

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```
import cv2
import matplotlib.pyplot as plt
import numpy as np
from google.colab.patches import cv2_imshow
```

✓ Getting the image

Will be using an image of Ray Chen

[Link to site](#)

[Link to image](#)

[Link to why I need the patch for cv2](#)

```
!curl -o rayChenImage.jpg https://www.bowdoinfestival.org/contents/media/2020/06/ray-topcover-10-square-400x400.jpg
#use cv2
image = cv2.imread('rayChenImage.jpg', cv2.IMREAD_UNCHANGED)

#show the image
#cv2.imshow('image', image)
#colab throws an error for that, so use patched version
cv2_imshow(image)
```

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✓ Image Flip

Easiest way of doing this is just swapping the columns. This can be done since this is a numpy array.

Credits: [link](#)

```
#flip
image = np.array([list(reversed(row)) for row in image])
cv2_imshow(image)
```



▼ Convolutional



```
#making the kernel
kernelEdge = np.array([[0, 1, 0],
                       [1, -4, 1],
                       [0, 1, 0]])

#applying the kernel filter
filtImg = cv2.filter2D(image, -1, kernelEdge)

#show the image
cv2.imshow(filtImg)
```

